
Python Pattern Questions Using Loops (20 Problems)

□ BASIC LEVEL (1–7)

1. Solid Square Star Pattern

Problem Statement:

Write a Python program to print a square pattern of stars of size N.

```
N = 4
****
****
****
****
```

2. Left-Aligned Right Triangle

Problem Statement:

Print a left-aligned right-angled triangle using stars.

```
N = 5
*
**
***
****
*****
```

3. Inverted Left-Aligned Triangle

Problem Statement:

Print an inverted left-aligned triangle of stars.

```
N = 5
*****
****
***
**
*
```

4. Single Row Number Pattern

Problem Statement:

Print numbers from 1 to N in a single row using loops.

```
N = 5  
12345
```

5. Right Triangle with Numbers

Problem Statement:

Print a right-angled triangle using increasing numbers in each row.

```
N = 4  
1  
12  
123  
1234
```

6. Repeated Number Triangle

Problem Statement:

Print a triangle where each row contains the same number.

```
N = 4  
1  
22  
333  
4444
```

7. Alphabet Triangle

Problem Statement:

Print a right-angled triangle using alphabets starting from A.

```
N = 4  
A  
AB  
ABC  
ABCD
```

☐ INTERMEDIATE LEVEL (8–14)

8. Floyd's Triangle

Problem Statement:

Print Floyd's Triangle using numbers.

```
N = 4
1
2 3
4 5 6
7 8 9 10
```

9. Centered Star Pyramid

Problem Statement:

Print a centered pyramid star pattern.

```
N = 4
  *
 * *
* * *
* * * *
* * * * *
```

10. Inverted Centered Pyramid

Problem Statement:

Print an inverted centered pyramid using stars.

```
N = 4
* * * * *
 * * * *
  * * *
   *

```

11. Number Pyramid

Problem Statement:

Print a centered number pyramid.

```
N = 4
  1
 1 2 1
1 2 3 2 1
1 2 3 4 3 2 1
```

12. Hollow Square Pattern

Problem Statement:

Print a hollow square of stars.

```
N = 5
*****
*       *
*       *
*       *
*****
```

13. Hollow Right Triangle

Problem Statement:

Print a hollow right-angled triangle using stars.

```
N = 5
*
**
* *
*  *
*****
```

14. Reverse Number Triangle

Problem Statement:

Print a triangle with decreasing numbers.

```
N = 4
1234
123
12
1
```

☐ ADVANCED LEVEL (15–20)

15. Diamond Star Pattern

Problem Statement:

Print a diamond shape using stars.

```
N = 4
  *
 * *
***
*****
*****
 * *
  *
```

*

16. Hollow Diamond Pattern

Problem Statement:

Print a hollow diamond using stars.

17. Butterfly Pattern

Problem Statement:

Print a butterfly star pattern using loops.

```
N = 4
*      *
**     **
***    ***
*****
***    ***
**     **
*      *
```

18. Pascal's Triangle

Problem Statement:

Print Pascal's Triangle up to N rows.

```
N = 5
    1
   1 1
  1 2 1
 1 3 3 1
1 4 6 4 1
```

19. Zig-Zag Star Pattern

Problem Statement:

Print a zig-zag pattern using stars.

```
N = 9
*   *   *
* * * *
  *   *
```

20. Hourglass Star Pattern

Problem Statement:

Print an hourglass pattern using stars.

```
N = 4
* * * * *
 * * * *
  * * *
   *
  * * *
 * * * *
* * * * *
```
